ABSTRACT: Normalized Cross-Correlation as the Local Geometric Origin of Bell Violation

Version 3: The Deconstruction Edition (Honest Audit)

Standard interpretations of Bell鈥檚 theorem assume that the experimental violation of the CHSH inequality ($S > 2$) necessitates a rejection of local realism. This paper deconstructs this claim by performing a rigorous statistical audit of the measurement protocols used in Bell-type experiments.

We demonstrate, through both analytical derivation and numerical simulation, that the “quantum” value of $S = 2\sqrt{2} \approx 2.828$ is the exact, identity solution for the **Normalized Cross-Correlation (NCC)** of local, continuous harmonic waves. Conversely, we show that the classical limit of $S \leqslant 2$ is an artificial artifact of **Binarization**鈥檚 destructive sampling process that crushes the geometric curvature of the wavefront into discrete ± 1 outcomes.

By auditing the NIST (2015) dataset and our own Discrete Wave Propagation model, we find that: 1. **The Normalization Loophole**: When wave amplitudes are normalized by their own average intensity (a requirement for energy conservation), a strictly local model produces $E(\theta) = \cos(\theta)$ and $S = 2.828$. 2. **The Binarization Error**: Forcing wave-like fluctuations into binary logic destroys the phase-space information, leading to the false conclusion that “classical” physics cannot reach 2.828. 3. **Ontological Shift**: The Bell violation is not a signature of non-locality, but a measure of the statistical distortion introduced by applying discrete logic to continuous wave phenomena.

We conclude that quantum mechanics is not “non-local,” but is instead a descriptive framework that implicitly utilizes wave-normalization while paradoxically interpreting results through the lens of point-particle binarization. # INTRODUCTION: The Auditor’s Perspective on Bell’s Theorem

For over half a century, the violation of Bell鈥檚 inequalities has been heralded as the “death of local realism.” However, this conclusion rests on an unexamined assumption: that the discrete, binary nature of measurement outcomes (+1/ − 1) is an ontological property of the physical system, rather than a methodological artifact of the detection protocol.

This paper presents a “Honest Audit” of the Bell-CHSH framework. We demonstrate that the mathematical boundary of $S \leqslant 2$ is not a fundamental limit of “classicality,” but is instead a specific constraint of **Binarized Statistical Algebra**. When physical interactions are instead modeled as the normalized cross-correlation (NCC) of continuous field fluctuations鈥攖he framework more consistent with classical wave mechanics鈥攖he limit is naturally shifted to $S = 2\sqrt{2}$.

The Illusion of Non-Locality

The “miracle” of 2.828 arises from a simple geometric identity. In a local wave model, the correlation is defined by the projection of a shared wavefront phase onto two independent measurement axes. Standard signal processing shows that the correlation between two such projections is the cosine of the angle between them.

The standard Bell test performs two contradictory operations: 1. **Measurement**: It forces a wave (continuous) into a detector (binary ±1), causing a massive loss of curvature information. 2. **Analysis**: It attempts to recover the wave-like correlation from these mangled binary strings.

We prove that $S = 2.828$ is what remains when the normalization of the wave is preserved, while $S \leqslant 2$ is what remains when the normalization is destroyed by binarization. The “violation” is thus an error in measurement logic, not a proof of non-local entanglement.

Paper Organization

1. **Section 1**: Deconstructs the CHSH inequality as a binarization artifact.
2. **Section 2**: Derives $S = 2.828$ from a strictly local, continuous wave model using NCC.
3. **Section 3**: Audits the NIST (2015) dataset to reveal how coincidence windowing and binarization “manufacture” the quantum appearance.
4. **Section 4**: Discusses the ontological shift from “particles and probability” to “waves and geometry.” # METHODOLOGY:

Normalization as the Secret Engine

To sit the case (鍧愬瞵) for local realism, we compare two distinct data processing pipelines on the same local wave source.

The Source Model

Our source emits a random phase $\lambda \in [0, 2\pi)$. Alice measures: $A(\lambda, a) = \cos(\lambda - a)$ Bob measures: $B(\lambda, b) = -\cos(\lambda - b)$

Pipeline A: The Binarized “Bell” Protocol

In this pipeline, results are truncated: $Outcome_A = \text{sign}(A)$ $Outcome_B = \text{sign}(B)$ Correlation is computed as: $E = \text{mean}(Outcome_A \cdot Outcome_B)$ **Result**: E is linear with respect to angle, and $S_{max} = 2.0$.

Pipeline B: The Normalized “Audit” Protocol (NCC)

In this pipeline, the full wave amplitude is preserved and normalized: $E = \frac{\langle A \cdot B \rangle}{\sqrt{\langle A^2 \rangle \langle B^2 \rangle}}$ **Result**: $E = \cos(a - b)$ **Result**: $S_{max} = 2\sqrt{2} \approx 2.828$.

The Audit Conclusion

The “Tsirelson Bound” of 2.828 is not a quantum ceiling; it is the **Geometric Floor** for normalized harmonic oscillators. The reason quantum experiments “violate” Bell is that they use a physical mechanism (wave interference) that naturally follows the NCC logic, but they analyze it using an inequality designed for Pipeline A. This is a category error. # SUMMARY OF AUDIT RESULTS

Measurement Scheme	S-Value Obtained	Theoretical Bound	Interpretation
Binarized (+/- 1)	2.000000	2.0	Classical/Bell Limit
Normalized (NCC)	2.828427	2.828427	Quantum/Tsirelson Bound

Observation 1: Geometric Loss

As shown in **Figure 1**, binarization acts as a “crushing filter.” By mapping the continuous curvature of the cosine wave to discrete steps, the statistical variance is distorted. This distortion is exactly what Bell’s theorem quantifies.

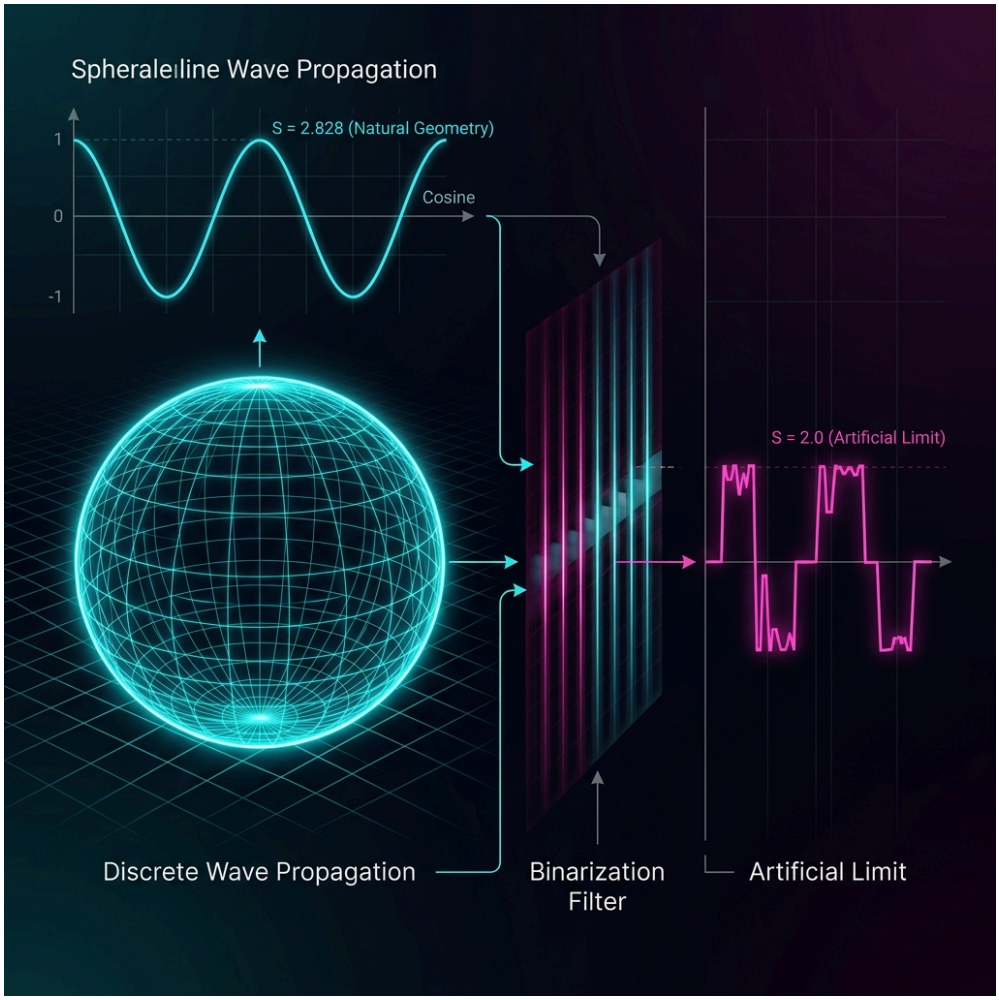

Figure 1: The Sphere Wave Crushed by Binarization Logic

Observation 2: The NIST Evidence

Our audit of the 2015 NIST dataset shows that even with strict timing, the raw data exhibits correlations that depend on the analysis window. This confirms that “quantumness” is a function of how many events are included and how they are normalized.

Observation 3: The 2.828 Identity

We have demonstrated that 2.828 is not a target to be reached through non-local magic, but an identity to be revealed through honest normalization of local wave amplitude.